

Q1.2.

$$\rightarrow e_i = \hat{y}_i - y_i$$

$$\text{Sample 1: } \rightarrow e_1 = 0 - 2 = -2$$

$$\text{Sample 2: } \rightarrow e_2 = 0 - 3 = -3$$

$$\frac{\partial L}{\partial w_1} = \frac{2}{N} \sum e_i x_{1i} \quad \frac{\partial L}{\partial w_2} = \frac{2}{N} \sum e_i x_{2i} \quad \frac{\partial L}{\partial b} = \frac{2}{N} \sum e_i$$

$$\frac{\partial L}{\partial w_1} = \frac{2}{2} [(-2)(1) + (-3)(2)] = -8 //$$

$$\frac{\partial L}{\partial w_2} = \frac{2}{2} [(-2)(1) + (-3)(1)] = -5 //$$

$$\frac{\partial L}{\partial b} = \frac{2}{2} [(-2) + (-3)] = -5 //$$